RODDY / LINK COMMANDER

Thin FDM cases - first-print guide

4 November 2026. Download includes both current designs and a cable alternative for the small dongle. These replace the older Rev E/F common enclosure.

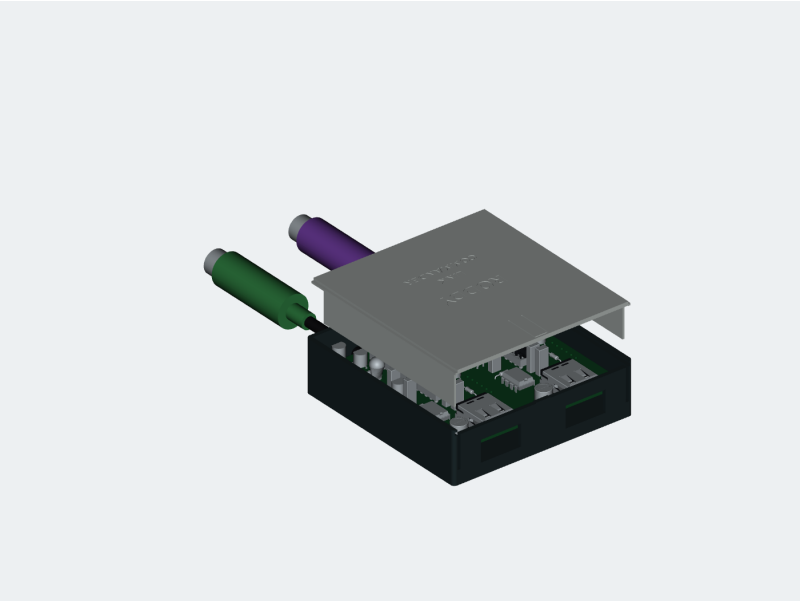


SMT Rev I dongle: 24.2 x 56.1 x 16.4 mm with captured plug; 24.2 x 49.3 x 16.4 mm with cable.
Through-hole Rev H dual: 76.2 x 79.3 x 21.4 mm. Dimensions exclude projecting plugs and cables.

Print: use the base-PRINT.stl and lid-PRINT.stl files, already oriented for the bed. Start with PETG, a 0.4 mm nozzle, 0.2 mm layers, four perimeters and 25-35% infill. Main walls, floor and lid are 1.6 mm. Print the slide-snap fit-test pair first. Do not scale the whole case.

CAD checks passed: valid case solids, manifold print meshes and sampled rigid lid clearance. Printed fit, snap force, fatigue and connector retention still need testing. Hardware and firmware are in development; no electronics are included.

Assembly and connector fit



Board retention: insert the USB end first, then lower the PCB onto the edge ledges. Slide the lid in from the USB end until its roof latch clicks. The lid skirts retain the board. Gently lift the latch tip to release; do not press it down onto the USB socket.

Direct dongle: the printed nose captures a real six-pin mini-DIN male insert, connected to the PCB by four short insulated wires. Nominal mating shield diameter is 9.5 mm, with 6.5 mm exposed beyond the case. The provisional internal insert envelope is 12 x 9 x 11 mm. A complete 50 mm cable plug does not fit this pocket intact. Check your insert and revise the holder if needed.

Cable versions: nominal 4.0 mm cable jackets pass through 4.4 mm exits. Internal cable-tie anchors provide jacket strain relief. Confirm retention with the actual cable and tie. The dual uses separate keyboard and mouse leads.

Stacked sockets: two aligned dongle cases need about 17 mm vertical centre spacing including a 0.6 mm gap. This is a case requirement, not a measured X16 specification. Use the included height gauge before assembly, or choose the short-cable variant.

Included: STL print parts, editable STEP case geometry, assembly previews, fit coupons and detailed READ_ME.txt instructions. Preview electronics and connector shapes illustrate fit; they are not manufacturing drawings or verified purchased connector models.